

Bipartite Graph

Difficulty: **Medium** Accuracy: **31.25%** Submissions: **219K** Points: **4** Average Time: **15m**

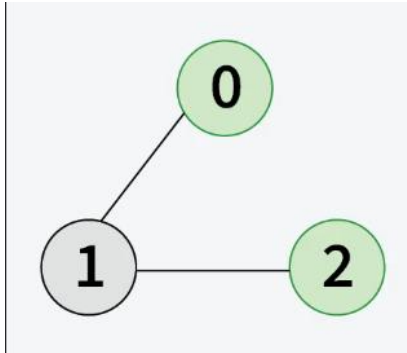
Given a Graph with **V** vertices (Numbered from **0** to **V-1**) and **E** edges. Check whether the graph is **bipartite** or not.

A [bipartite graph](#) can be colored with two colors such that **no two adjacent vertices** share the **same color**. This means we can divide the graph's vertices into two distinct sets where:

- All edges connect vertices from one set to vertices in the other set.
- No edges exist between vertices within the same set.

Examples:

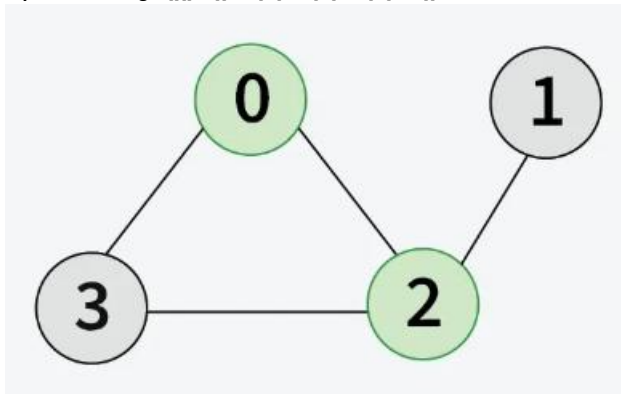
Input: $V = 3$, $\text{edges}[][] = [[0, 1], [1, 2]]$



Output: true

Explanation: The given graph can be colored in two colors so, it is a bipartite graph.

Input: $V = 4$, $\text{edges}[][] = [[0, 3], [1, 2], [3, 2], [0, 2]]$



Output: false

Explanation: The given graph cannot be colored in two colors such that color of adjacent vertices differs.